Summary in Graph

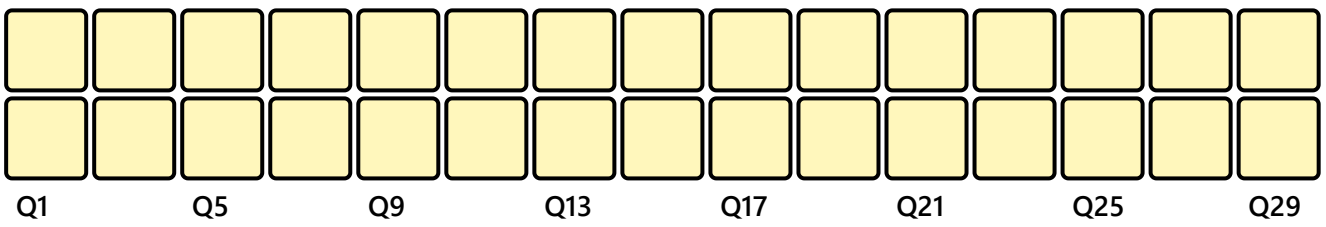
Exam Summary (GATE Overflow CS Test Series 2026 | Mix Subjects | Test 3)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



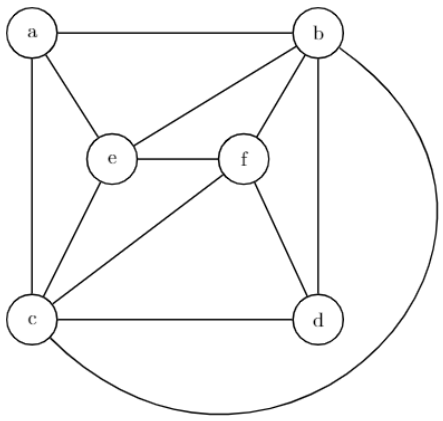
* Average Stats based on first-attempt stats of Top 10% (2) out of 23 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Numerical Type | Award: 1 | Penalty: 0 | Graph Theory

The chromatic number of the following graph is _____



Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 Multiple Select Type | Award: 1 | Penalty: 0 | Theory of Computation

The regular expression for the language having exclusively the strings, starting and ending with different symbols, over $\Sigma = \{a, b\}$ is given by _____ (Mark all the appropriate choices)

A. $a(a+b)^*b + b(a+b)^*b$
B. $a(a+b)^*b + a(a+b)^*a$
C. $b(a+b)^*a + a(a+b)^*a$
D. $a(a+b)^*b + b(a+b)^*a$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

In a binomial distribution, if n is the number of trials and p is the probability of success, then the mean value is given by _____

A. np
B. n
C. p
D. $np(1-p)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Multiple Select Type | Award: 1 | Penalty: 0 | Theory of Computation

A context-free grammar G is ambiguous if and only if

A. some string $w \in L(G)$ has different leftmost and rightmost derivations.
B. some string $w \in L(G)$ has at least two different parse trees.
C. every string $w \in L(G)$ has at least two different parse trees.
D. some string $w \in L(G)$ has at least two different leftmost derivations.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 Multiple Select Type | Award: 1 | Penalty: 0 | Databases

Which of the following is/are correct? (Mark all the appropriate options)

A. Every cascadeless schedule is a recoverable schedule.
B. Every recoverable schedule is a cascadeless schedule.
C. Every strict schedule is a cascadeless schedule.
D. Every cascadeless schedule is a strict schedule.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

Which of the following statements is/are correct? (Mark all the appropriate choices)

A. A cycle in the wait-for graph is a necessary condition for deadlock to exist
B. A cycle in the wait-for graph is a sufficient condition for deadlock to exist when only single instance is available per resource
C. Testing for view serializability is NP-complete
D. Testing for conflict serializability is NP-complete

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Match the following:

a.	Transaction Failure	i.	Transaction Manager
b.	Concurrency Control	ii.	Buffer Manager
c.	Data transfer between disk and memory	iii.	Scheduler
d.	ACID Properties	iv.	Recovery Manager

A. $a-i, b-iii, c-ii, d-iv$
B. $a-iv, b-iii, c-ii, d-i$
C. $a-ii, b-iii, c-iv, d-i$
D. $a-iii, b-iv, c-ii, d-i$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Consider a situation for a transaction T where a later transaction has already updated the value of the item, and the value that the older transaction is writing is based on an obsolete value of the item. In this case as per Thomas' write rule,

A. the write operation can safely be ignored
B. only transaction T must be rolled back
C. both transaction T and the later transaction must be rolled back
D. only the later transaction must be rolled back

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

In which of the following cases the order of execution is important? (Mark all the appropriate choices)

A. If two transactions only read a data item
B. If two transactions either read or write completely separate data items
C. If one transaction writes a data item and another either reads the same data item
D. If one transaction writes a data item and another either writes the same data item

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

Which of the following statements is/are correct? (Mark all the appropriate choices)

A. With strict $2\mathrm{PL}$, transactions can be serialized in the order in which they commit
B. Strict $2\mathrm{PL}$, holds only exclusive locks until the end of the transaction
C. Rigorous $2\mathrm{PL}$, holds both shared and exclusive locks until the end of the transaction
D. With rigorous $2\mathrm{PL}$, transactions can be serialized in the order in which they commit

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider a B^+- tree in which the search key is 24 bytes long, block size is 2048 bytes, recorder pointer is 20 bytes long and the block pointer is 16 bytes long. If the maximum number of keys that can be accommodated in each leaf node is $'a'$ and that in each internal node (non-leaf) is $'b'$, the value of $b-a$ is _____

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Dr. Unnikrishnan did a UPI transaction to book a bus ticket on his mobile phone and as soon as he received a success confirmation his phone battery died and the phone tuned off. If that was the last ticket available and many others were simultaneously trying for that ticket, which of the following is/are a possible outcome? (Mark all the appropriate options)

A. The transaction is cancelled and Doctor gets his refund
B. Doctor loses his money and ticket gets assigned to someone else
C. Doctor gets his ticket when he restarts his mobile phone
D. Both doctor and someone else gets assigned the ticket and after contacting the customer care one of their ticket gets cancelled and he gets a refund

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

If there are n transactions each having $m_1, m_2, \ldots, m_n$ operations respectively, the number of concurrent schedules possible is _____ (Mark all the appropriate options)

A. $\dfrac{(m_1+m_2+m_3+\cdots+m_n)!}{m_1!m_2!m_3!\ldots m_n!}$
B. $n! - \dfrac{(m_1+m_2+m_3+\cdots+m_n)!}{m_1!m_2!m_3!\ldots m_n!}$
C. ${}^{m_1+m_2+m_3+\ldots+m_n}C_{m_1} \times {}^{m_2+m_3+\ldots+m_n}C_{m_2} \times {}^{m_3+\ldots+m_n}C_{m_3} \times \ldots \times {}^{m_n}C_{m_n}$

D. $\dfrac{(m_1 \times m_2 \times m_3 \times \cdots \times m_n)!}{m_1!m_2!m_3!\ldots m_n!}$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider 4 transactions T_1, T_2, T_3 and T_4 having $1, 4, 2$ and 3 operations respectively. If a is the total number of serial schedules and b is the total number of concurrent schedules then the total number of non-serial schedules, $c = b - a =$ _____

Your Answer: | Correct Answer: 12576 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Which of the following is/are correct? (Mark all the appropriate options)

A. In a Serial schedule, a transaction is executed completely before starting the execution of another transaction.
B. In a Strict schedule, if the write operation of a transaction precedes a conflicting operation (Read or Write operation) of another transaction then the commit or abort operation of such transaction should also precede the conflicting operation of other transaction.
C. In a Cascadeless schedule, if a transaction is going to perform read operation on a value, it has to wait until the transaction which is performing write on that value commits.
D. In a Recoverable schedule, if a transaction is reading a value which has been updated by some other transaction then this transaction can commit only after the commit of other transaction which is updating the value.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider the following schedules

First Schedule

T_1	T_2	T_3
$R(A)$		
		$R(A)$
$W(A)$		
	$R(A)$	
	$W(A)$	

If the above schedule is conflict serializable, then let a be the number of topological orderings of the precedence graph of the schedule. Otherwise, let $a = 0$.

Second Schedule

T_1	T_2	T_3	T_4
			$R(A)$
	$R(A)$		
		$R(A)$	
$W(B)$			
	$W(A)$		
		$R(B)$	
	$W(B)$		

If the above schedule is conflict serializable, then let b be the number of topological orderings of the precedence graph of the schedule. Otherwise, let $b = 0$. The value of $a + b + (a \times b) =$ _____

Your Answer: | Correct Answer: 7 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider the following schedule

Schedule

T_1	T_2	T_3	T_4
$R(x)$			
	$R(x)$		
		$R(x)$	
			$R(x)$
$W(y)$			
	$W(y)$		
		$W(y)$	
			$W(y)$

If the number of view equivalent serial schedules is equal to a, the value of $2^a =$ _____

Your Answer: | Correct Answer: 64 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Which of the following is/are correct with respect to 2-phase locking $(2\mathrm{PL})$? (Mark all the appropriate options)

A. $2\mathrm{PL}$ always ensures serializability, but cascading rollbacks, deadlocks and starvation are possible
B. Strict $2\mathrm{PL}$ ensures recoverability and cascadelessness but still deadlock is possible
C. With strict $2\mathrm{PL}$, transactions can be serialized in the order in which they commit
D. Every conservative $2\mathrm{PL}$ is rigorous $2\mathrm{PL}$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Which of the following is/are correct with respect to concurrency control of database transactions? (Mark all the appropriate options)

A. There are schedules common to both $2\mathrm{PL}$ and time-stamp based protocol
B. Basic time-stamp based concurrency control ensures serializability and freedom from deadlock but the schedule may not be recoverable

C. The use of Thomas' write rule allows schedules to be generated that would not have been possible with basic timestamp protocol

D. There are schedules that can be produced by basic time-stamp based protocol but not $2PL$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 Numerical Type | Award: 2 | Penalty: 0 | Probability

Assume that a light bulb lasts on average 100 hours. Assuming exponential distribution, the probability that it lasts either more than 200 hours or not more than 50 hours is _____. (Rounded to 2 decimal points)

Your Answer: | Correct Answer: 0.52:0.53 | Not Attempted | Time taken: 00min 00sec | Discuss

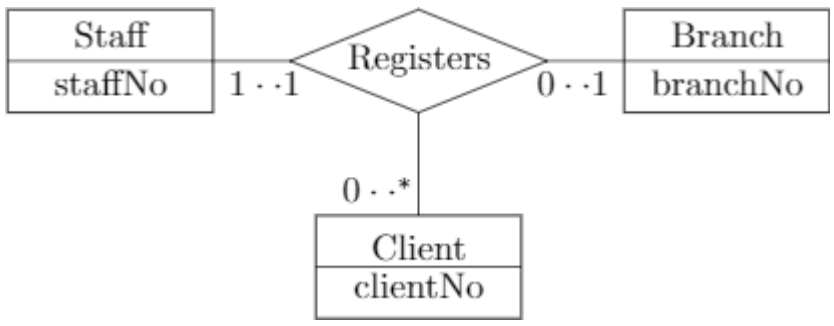
Q #21 Numerical Type | Award: 2 | Penalty: 0 | Databases

The numbers of relations required when the below ER diagram is converted to Relational model is _____

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 Numerical Type | Award: 2 | Penalty: 0 | Databases

The numbers of relations required when the below ER diagram is converted to Relational model is _____



Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 Numerical Type | Award: 2 | Penalty: 0 | Databases

Given that the relations $R(A, B)$ and $S(C, D)$.

R

A	B
1	a
2	c

S

C	D
1	b
3	d

If the number of rows in $R \bowtie_{\langle A=C \rangle} S = a$(natural join), the number of rows in $R ⟕_{\langle A=C \rangle} S = b$ (left outer join), the number of rows in $R ⟖_{\langle A=C \rangle} S = c$ (right outer join), and the number of rows in $R ⟗_{\langle A=C \rangle} S = d$ (full outer join), then the value of $2a + 3b + 5c + 7d =$ _____

Your Answer: | Correct Answer: 39 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Given the Sailors table

Sailors

Sid	Sname	Rating	Age
1	Peter	10	23
2	Alex	9	24
3	Alex	9	25
4	Michael	9	24

What does the $\Pi_{\text{Age}}\left[\sigma_{\text{Rating}=9}[\text{Sailors}]\right]$ give?

A.

Age
24
25
24

B.

Age
23
24
25

C.

Age
23
24

D.

Age
24
25

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 03sec | Discuss

Q #25 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following SQL query:

```
SELECT *
FROM Staff s, Branch b
WHERE s.branchNo = b.branchNo AND
(s.position = 'Manager' AND b.city = 'Bangalore');
```

Which of the following relational algebra query gives the most optimal execution strategy for the above SQL query assuming no indexing and that there are 1000 tuples in Staff, 100 tuples in Branch, 100 Managers (one for each branch), and 10 Bangalore branches?

A. $\sigma_{(position='Manager')\wedge(city='Bangalore')\wedge(\text{Staff}.branchNo=\text{Branch}.branchNo)}(\text{Staff} \times \text{Branch})$

B. $\sigma_{(position='Manager')\wedge(city='Bangalore')}(\text{Staff} \bowtie_{\text{Staff}.branchNo=\text{Branch}.branchNo} \text{Branch})$

C. $\sigma_{(position='Manager')}(\text{Staff} \bowtie_{\text{Staff}.branchNo=\text{Branch}.branchNo} \text{Branch})$

D. $(\sigma_{position='Manager'}(\text{Staff})) \bowtie_{\text{Staff}.branchNo=\text{Branch}.branchNo} (\sigma_{city='Bangalore'}(\text{Branch}))$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

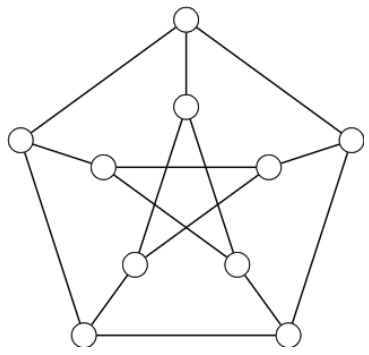
Which of the following languages is/are regular? (Mark all the appropriate choices)

A. Set of binary strings containing equal number of " 00 " and " 11 "
B. Set of binary strings containing equal number of " 01 " and " 10 "
C. Set of binary strings containing equal number of " 01 " and " 11 "
D. Set of binary strings containing $1,000,000$ number of " 11 "

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 Numerical Type | Award: 2 | Penalty: 0 | Graph Theory

The Chromatic number of the following graph is _____



Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 Numerical Type | Award: 2 | Penalty: 0 | Graph Theory

The sum of the number of vertices and edges of a graph with degree sequence $(6,5,4,3,3,2,1)$ is _____

Your Answer: | Correct Answer: 19 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 Multiple Select Type | Award: 2 | Penalty: 0 | Graph Theory

Which of the following is/are possible for unlabelled trees/graphs? (Mark all the appropriate choices)

A. Two different trees with the same number of vertices and the same number of edges.
B. Two different simple graphs with 8 vertices all of degree 2.
C. Two different simple graphs with 5 vertices all of degree 4.
D. Two different graphs with 7 vertices all of degree 3.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

The language accepted by a Turing Machine if its Read/Write head is converted to a Read Only one is?

A. Recursive but not necessarily context-free
B. Deterministic context-free but not necessarily regular
C. Regular
D. None of the above

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

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